# Tutorial 5

## MAT3007 Optimization, 2023 Summer

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# Outline

1. Simplex Tableau
2. Degeneracy
3. Two-Phase Simplex Method
4. Big-M Method

Simplex method:

the idea: neighboring BFS : 2 BS differ by one basic index

$m(n-m)$ potential neighbors

↓

choose which one change

Select a non-basic variable $x_j$ to enter the basis

stopping criterion $\bar{c}_j \geq 0$

There are cases that some variables in a BFS $= 0$ degeneracy

Simpler implementation of simplex method → avoid matrix inversion!

# Review: Simplex Tableau

The simplex tableau is a table with the following structure when the basis is $B$ (the corresponding basis matrix and objective coefficients are $A_B$ and $\mathbf{c}_B$):

$Ax=b \to A_B^{-1}Ax = A_B^{-1}b$

$x=[x_b; x_N] = [A_B^{-1}b; 0]$

| | |
|---|---|
| $\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A$ | $-\mathbf{c}_B^T A_B^{-1}\mathbf{b}$ |
| $A_B^{-1}A$ | $A_B^{-1}\mathbf{b}$ |

$-\mathbf{c}_B^T A_B^{-1}\mathbf{b}$ → $-$(objective value) $= -c_B^T x_B$

$A_B^{-1}\mathbf{b}$ → $x_B$

reduced cost

- The lower part is simply a transformation of the constraints. In particular, the right hand side column is the current basic variables.
- The upper left part is the reduced costs, and the upper right corner is the negative of the objective value.

$A_B^{-1}[A_B; A_N] = [I_m \quad A_B^{-1}A_N]$

Therefore, The simplex tableau should look like (after reordering the columns)

Reduced costs for basic variables are zero.

| | | |
|---|---|---|
| $\mathbf{0}_m$ | $\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A_N$ | $-\mathbf{c}_B^T \mathbf{x}_B$ |
| $\mathbf{I}_m$ | $A_B^{-1} A_N$ | $\mathbf{x}_B$ |

⇒ "canonical form" of LP - easy to see whether BFS optimal.

Here $\mathbf{0}_m$ is a vector of $m$ zeros and $\mathbf{I}_m$ is the $m$-dimensional identity matrix.

## From Simplex method to simplex tableau:

1. compute reduced cost: $\bar{c}_j = c_j - c_B^T A_B^{-1} A_j$

2. choosing incoming basic index $\bar{c}_j < 0$

3. Compute $\theta^* = \min_{i \in B, d_i < 0} \left\{ -\frac{x_i}{d_i} \right\}$ $\quad l = \arg\min_{l \in B, d_l < 0} \left\{ \frac{x_i}{d_i} \right\}$

   . choosing outgoing basix index $\cdot\ l$ .

4. update with new basis. $y = x + \theta^* d$ .

$x + \theta^* d$

The simplex tableau will maintain a canonical form for each iteration until we reach optimality. Each iteration consists of

- Determine optimality: look at the top row (the reduced costs)
  - If all nonnegative, then the solution is already optimal
  - Otherwise, choose the smallest index $j$ such that $\overline{c}_j < 0$ to be the incoming basis (pivot column)
- Perform the minimal ratio test (between the current basic variables and the incoming column — positive entries only).
- Choose the row that attains the smallest ratio to be the pivot row, and determine the outgoing basis (smallest index rule)
- Update the tableau
  - Divide each entry in the pivot row by the pivot element.
  - Add proper multiples of the new pivot row to each other rows, such that all other elements in the pivot column become zeros (including the top row).

Recall:

- Pivoting: transform from one canonical form to another canonical form.
- Minimal Ratio test (MRT)

  step size $\theta^* = \min_{d_i < 0,\, i \in B} \frac{-x_i}{d_i}$

  $x_i$ : $i$th entry of basic solution. $d_B = -A_B^{-1} A_j$

  lower right part "$x_B$"

In simplex tableau:

$$\theta^* = \min_i \left\{ \frac{x_B}{\bar{A}_{ij}} , \bar{A}_{ij} > 0 \right\}$$

lower left part "$A_B^{-1} A_N$"

# Exercise 1

Consider the following tableau

| $B$ | $\beta$ | 0 | 0 | 0 | $\delta$ | 2 | 0 |
|---|---|---|---|---|---|---|---|
| 2 | $-1$ | 1 | 0 | 0 | $\eta$ | 0 | 1 |
| 4 | $-1$ | 0 | 0 | 1 | $-1$ | 1 | 2 |
| 3 | $\alpha$ | 0 | 1 | 0 | 1 | 1 | $\gamma$ |

The entries of $\alpha, \beta, \eta, \delta$ and $\gamma$ are unknown parameters. Find sufficient conditions on the range of them such that the following statements are true.

1. The current basic solution is optimal for the problem
2. The current basic solution is feasible. At the next iteration, $x_5$ is the only candidate to enter the basis. And if $x_5$ enters the basis, $x_2$ leaves the basis according to Bland's rule.
3. The current basic solution is feasible, but the problem is unbounded.
4. The last row indicates that the problem is infeasible.

1. $\beta \geq 0$ $\quad \delta \geq 0$ $\quad \gamma \geq 0$

(the reduced cost non-negative ; $\gamma = x_B \geq 0$ feasible)

2. $x_5$ is only candidate as incoming basis:

only $\delta < 0$ $\quad \beta \geq 0$.

$\frac{1}{j} = \min\left\{\frac{1}{j}, \frac{\gamma}{1}\right\}$.

$x_2$ outcoming according to Bland's rule:

$\frac{1}{j} \leq \gamma$

$j > 0$

Use smallest index rule for choosing both incoming & outcoming basis.

- smallest index rule: choose smallest $j$ when $\bar{c}_j < 0$

$d_i = -A_B^{-1} A_j < 0$

$\frac{1}{j} \leq \frac{\gamma}{1} = \gamma$.

$\because -1 \times$

$\therefore j > 0$ is OK!

3. feasible: $\gamma \geq 0$ $\quad d_i > 0$

unbounded: $\beta < 0$. ← sufficient, go into 1st col.

$\frac{\gamma}{d} < 0 \rightarrow d \leq 0$

4. last row → infeasible:

"$x_B$"

$\gamma < 0$.

$\alpha \geq 0$

# Exercise 2

Use the simplex method to solve the following problem

$$\begin{aligned} \max \quad & 3x_1 + 4x_2 \\ s.t. \quad & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 5 \\ & x \geq 0 \end{aligned} \tag{1}$$

Standarlize:

$$\min \quad -3x_1 - 4x_2$$

$$\text{s.t.} \quad x_1 + x_2 + x_3 = 4$$
$$2x_1 + x_2 + x_4 = 5$$
$$x_1, x_2, x_3, x_4 \geq 0$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \qquad C = [-3, -4, 0, 0]$$

$$A_B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \qquad b = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

| | | |
|---|---|---|
| $c^T - c_B^T A_B^{-1} A$ | $-c_B^T A_B^{-1} b$ | $-(c$ |
| $A_B^{-1} A$ | $A_B^{-1} b$ | $\to x_B$ |

reduced cost.

| | | |
|---|---|---|
| $0_m$ | $c^T - c_B^T A_B^{-1} A_N$ | $-c_B^T x_B$ |
| $I_m$ | $A_B^{-1} A_N$ | $x_B$ |

## Starting tableau:

choose starting basis $B := \{3, 4\}$.

$$\therefore A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad A_N = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \qquad A_B^{-1} A_N = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$x_B = A_B^{-1} b = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \qquad C_B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$C^T - C_B^T A_B^{-1} A = C^T \qquad -C_B^T x_B = 0$$

top row

| B | $-3$ | $-4$ | $0$ | $0$ | $0$ |
|---|---|---|---|---|---|
| 3 | 1 | 1 | 1 | 0 | 4 |
| 4 | 2 | 1 | 0 | 1 | 5 |

row 1, row 2

By Bland's rule, choose smallest $j$ when $\bar{c}_j < 0$

in index: 1 — Pivot column is column 1

By MRT:

- Minimal Ratio test (MRT)

step size $\theta^* = \min_{d_i<0,\, i\in B} \frac{-x_i}{d_i}$

$x_i$ : ith entry of basic solution. $d_B = -A_B^{-1}A_j$

在 tableau 中. $\theta^* = \min_i \{ \frac{\bar{b}_i}{\bar{A}_{ij}} : \bar{A}_{ij} > 0 \}$

lower right — lower left

pivoting row

outgoing corresponding index (4)

B: 3, 4. $\min\{\frac{4}{1}, \frac{5}{2}\} = 2.5$

∴ 2th row is pivoting row.

| B | − |
|---|---|

$\times \frac{3}{2}$ of row 2 · add it on top row

$\times -\frac{1}{2}$ of row 2 add it to row 1

| B | | | | | |
|---|---|---|---|---|---|
| | 0 | $-\frac{5}{2}$ | 0 | $\frac{3}{2}$ | $5 \times \frac{3}{2} + 0 = \frac{15}{2}$ |
| 3 | 0 | $\frac{1}{2}$ | 1 | $-\frac{1}{2}$ | $4 - \frac{1}{2} \times 5 = \frac{3}{2}$ |
| 1 | 1 | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | $\frac{5}{2}$ |

$$\text{inv}(A_B) = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$b = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$A_B^{-1} A_N = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$X_B = A_B^{-1} b = \begin{bmatrix} 3 \\ \frac{1}{2} \\ \frac{5}{2} \end{bmatrix}$$

Reduced cost:

$$C_B = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$

$$-C_B^T \cdot X_B = \frac{15}{2}$$

$$C^T - C_B^T A_B^{-1} A_N =$$

| | | | | | |
|---|---|---|---|---|---|
| | | $-\frac{5}{2}$ | 0 | $\frac{3}{2}$ | $\frac{15}{2}$ |
| B | 0 | $-\frac{5}{2}$ | 0 | $\frac{3}{2}$ | $\frac{15}{2}$ |
| 3 | 0 | $\frac{1}{2}$ | 1 | $-\frac{1}{2}$ | $\frac{3}{2}$ |
| 1 | 1 | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | $\frac{5}{2}$ |

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# Review: Degenerate BFS and Pivoting Rules

$x = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix}$ $A = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 1 \end{bmatrix}$ $b = \begin{bmatrix} 8 \\ 4 \\ 4 \end{bmatrix}$

$x_B = A_B^{-1} b = (0, 4, 8)$

$\min_{\{i \in B, d_i < 0\}} \left(-\frac{x_i}{d_i}\right)$ ($d_i < 0$)

$x_1' = 0 \rightarrow$ cycle

**Definition 1 (Degeneracy).**

We call a basic feasible solution $x$ (non)degenerate if (none) some of the basic values are 0.

**Theorem 2 (Bland's Rule).**

*If we use smallest index rule for choosing both the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.*

outgoing.

**Smallest index rule:** Choose the smallest index $j$ with $\overline{c}_j < 0$

MRT

# Exercise 3

Use the simplex method to solve the following problem

$$
\begin{aligned}
\max \quad & z = 10x_1 - 57x_2 - 9x_3 - 24x_4 \\
\text{s.t.} \quad & 1/2x_1 - 11/2x_2 - 5/2x_3 + 9x_4 \leq 0 \\
& 1/2x_1 - 3/2x_2 - 1/2x_3 + x_4 \leq 0 \\
& x_1 + x_2 + x_3 + x_4 \quad \leq 1 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}
$$

Standardlize:

$$\min \quad -10x_1 + 57x_2 + 9x_3 + 24x_4$$

$$\text{s.t.} \quad \frac{1}{2}x_1 - \frac{11}{2}x_2 - \frac{5}{2}x_3 + 9x_4 + x_5 = 0$$

$$\frac{1}{2}x_1 - \frac{3}{2}x_2 - \frac{1}{2}x_3 + x_4 + x_6 = 0$$

$$x_1 + x_2 + x_3 + x_4 + x_7 = 1$$

$$x \geq 0$$

$$\underset{\sim}{c} = [-10, 57, 9, 24, 0, 0]$$

$$\underset{\sim}{A} = \begin{bmatrix} \frac{1}{2} & -\frac{11}{2} & -\frac{5}{2} & 9 & 1 & 0 & 0 \\ \frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

3

$m=3$

$$\underset{\sim}{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

good $A_B$.

iter 2: column 1. minimum = 0 ! $\min\left\{\frac{0}{\frac{1}{2}}, \frac{0}{\frac{1}{2}}, \frac{1}{1}\right\}$

add 20 x row 2 to the top row:

$x_1$ in $x_6$ out !

| B: | 0 | 27 | -1 | 44 | 0 | 20 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 0 | -4 | -2 | 8 | 1 | -1 | 0 | 0 |
| 1 | $\frac{1}{2}$ | $-\frac{3}{2}$ | $-\frac{1}{2}$ | 1 | 0 | 1 | 0 | 0 |
| 7 | 0 | 4 | 2 | -1 | 0 | -2 | 1 | 1 |

add 10 x row 3 to avoid entering row 1 now;
add -1 x row 3 to the first row.
add -2 x row 3 to the last row

x -t $x_3$ in.

iter 1: choose $B = \{5, 6, 7\}$

$$A_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad C_B^T = \begin{bmatrix} 24 \\ 0 \\ 0 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$x_B = A_B^{-1} b = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$ ← ∃ 0 ! → degenerate

$$A_N = \begin{bmatrix} \frac{1}{2} & -\frac{11}{2} & -\frac{5}{2} & 9 \\ \frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} = A_B^{-1} A_N .$$

$C_N^T = \cdots$

$-C_B^T x_B = 0$.

| row | B | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | B | −r 0 | 57 | 9 | 24 | 0 | 0 | 0 | 0 |
| 2 | 5 | $\frac{1}{2}$ | $-\frac{11}{2}$ | $-\frac{5}{2}$ | 9 | 1 | 0 | 0 | 0 |
| 3 | 6 | $\frac{1}{2}$ | $-\frac{3}{2}$ | $-\frac{1}{2}$ | 1 | 0 | 1 | 0 | 0 |
| 4 | 7 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |

0

↑

by bland's rule

index 1 → incoming index. $\bar{c}_j < 0$.

by bland's rule.

add $\underline{2}0$ x row2. add it to the top row.

$-1$ x row2 add it to the row 3

$-2$ x row2 add ... row 4.

$\times 2$. to make $\frac{1}{2}$ → 1

iteration 2 ---

optimal value

$\frac{1}{2}$

optimal

$[\frac{1}{2}, 0, \frac{1}{2}, 0, 5, 0, 0]^T$

7 lef

$x_6$: $A_B = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 1 & 1 \end{bmatrix}$ $A_B^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

5
1 → $A_B = \begin{bmatrix} 1 & \frac{1}{2} & -\frac{5}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 1 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix}$
3

$x_1$ left $x_3$ in.

no 6 now use 1.

B
5
1
7

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# Review: Two-Phase Method in Simplex Tableau

Finding initial is hard. $A_B = I_m$

**Phase I:**

- The auxiliary problem is not in a canonical form: Although the constraint contains an identity matrix (which is the initial basis), the corresponding objective coefficients are not 0's
- We need to compute the top row (reduced costs)
  - For basic part, the reduced costs are 0
  - For nonbasic part, $\bar{c}_j = c_j - \boldsymbol{c}_B^T A_B^{-1} A_j = -\boldsymbol{e}^T A_j$, so the $j$th reduced cost is the negative of the sum of that column
- This also applies to the initial objective value, which equals the negative of the sum of the right hand side vector.

$0$
$I_m$

## Phase I:

In the two-phase simplex method, we first solve an auxiliary problem (**1** is an all-one vector):

$$\begin{aligned} \text{minimize}_{x,y} \quad & \mathbf{1}^\top y := \textstyle\sum_{i=1}^n y_i \\ \text{subject to} \quad & Ax + y = b \\ & x, y \geq 0 \end{aligned}$$

Without loss of generality, we can assume $b \geq 0$ (otherwise, we pre-multiply that row by $-1$).

y

A

**Phase II:**

- Drop all the columns associated with the auxiliary variables
- Change the top row to the reduced costs corresponding to the original problem by using the reduced cost formula:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^T A_B^{-1} A_j$$

  Also compute the current objective value
- Continue the simplex tableau to solve the original problem

If the optimal solution to the auxiliary problem is degenerate, then replace the basic indices for the auxiliary variables with (any) indices of original variables and recalculate the table.

# Exercise 4: Two-phased Method

Use the two-phase simplex method to completely solve the linear optimization problem:

$$
\begin{aligned}
\min \quad & 2x_1 + 3x_2 + 3x_3 + x_4 - 2x_5 \\
s.t. \quad & x_1 + 3x_2 + \quad + 4x_4 + x_5 = 2 \\
& x_1 + 2x_2 + \quad - 3x_4 + x_5 = 2 \\
& x_1 + 4x_2 - 3x_3 + \quad + \quad = 1 \\
& x \geq 0
\end{aligned}
$$

$-/ \quad ( \qquad ) = (-1) \cdot (4)$

$$\min \quad y_1 + y_2 + y_3$$

$$\text{s.t.} \quad x_1 + 3x_2 + \cdots + 4x_4 + x_5 + y_1 + \qquad = 2$$

$$x_1 + 2x_2 + \cdots - 3x_4 + x_5 + y_2 \qquad = 2$$

$$-x_1 - 4x_2 + 3x_3 + \qquad + y_3 = 1$$

$$x, y \geq 0$$

$A_B \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

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# Review: Big-M Method

Solve LP without a starting BFS. Consider the following auxiliary problem:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^T\mathbf{x} + M\sum_{i=1}^{m} y_i \\ \text{subject to} \quad & A\mathbf{x} + \mathbf{y} = \mathbf{b} \\ & \mathbf{x}, \mathbf{y} \geq 0 \end{aligned}$$

This problem has an initial BFS $\mathbf{y} = \mathbf{b} \geq 0$ (again assuming $\mathbf{b} \geq 0$). Now we can use simplex to solve it. In the simplex procedure, pretend that $M$ is a very large value (larger than any specified number).

# Exercise 5: Big-M Method

Use the big-M simplex method to completely solve the linear optimization problem:

$$\begin{aligned} \min \quad & 4x_1 + x_2 + x_3 \\ s.t. \quad & 2x_1 + x_2 + 2x_3 = 4 \\ & 3x_1 + 3x_2 + x_3 = 3 \\ & x \geq 0 \end{aligned}$$

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